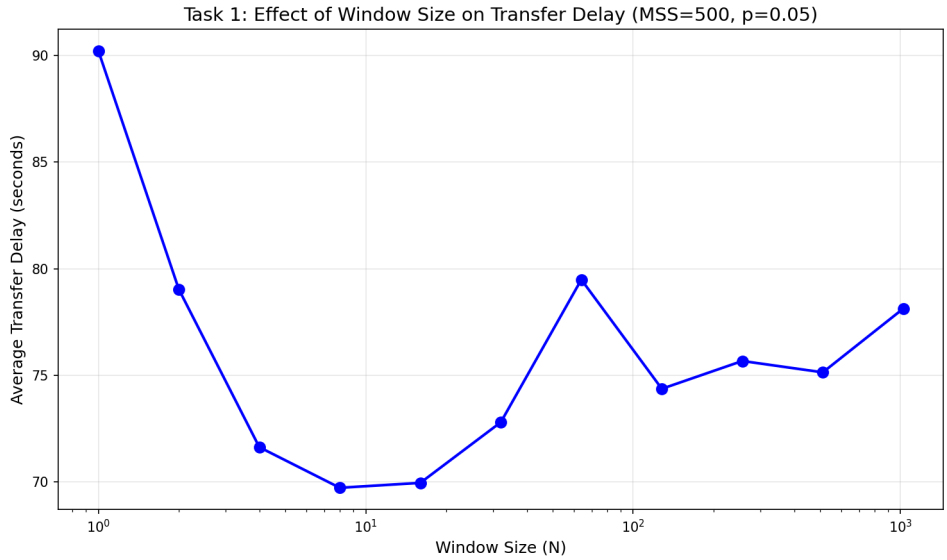
UDP File Transfer Results

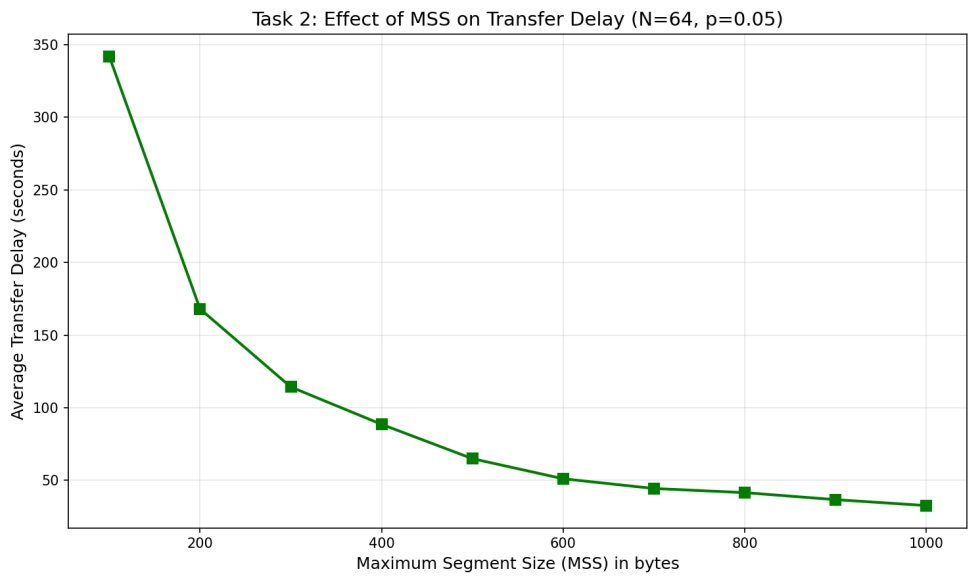
A. Go Back N Protocol

Task 1:



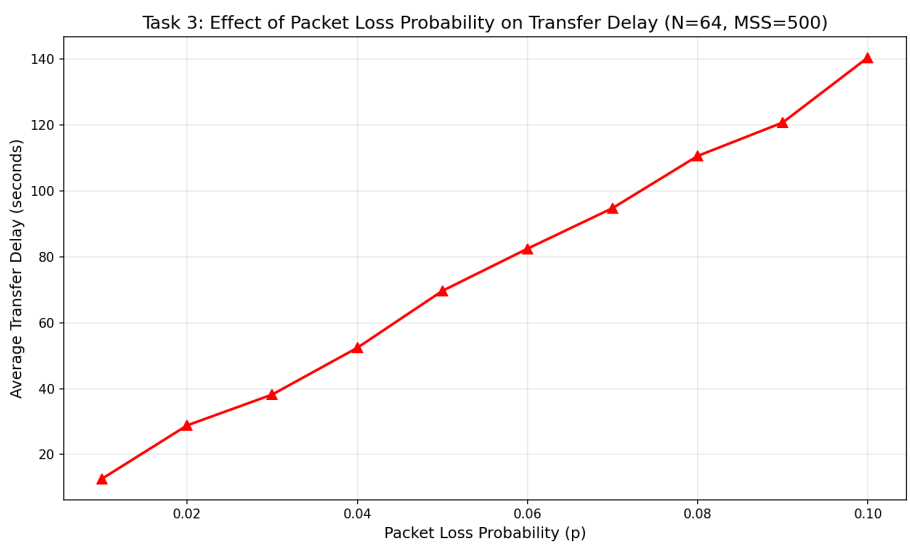
Interpretation: Increasing the window size reduces transfer delay at first, because the sender can keep more packets in flight and use the link more efficiently. The improvement is strongest going from very small windows to moderate ones, then the curve flattens and becomes a bit noisy, which suggests diminishing returns once the wind

Task 2:



Interpretation:Larger MSS values consistently lower the transfer delay. This is the clearest trend of the three plots: sending more data per packet reduces protocol overhead and the number of packets needed, so the file finishes much faster as MSS grows.

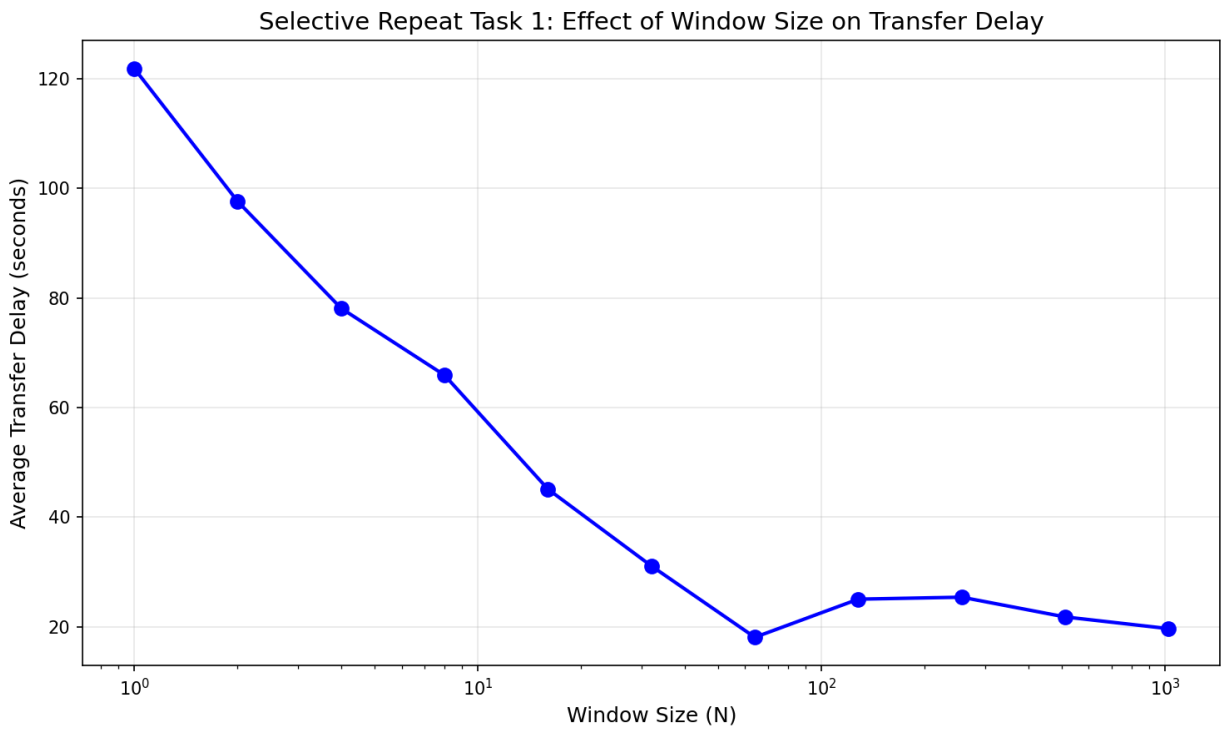
Task 3:



Interpretation:As packet loss probability increases, delay rises sharply. This matches Go-Back-N behavior: more loss means more timeouts, retransmissions, and wasted bandwidth, so the transfer becomes increasingly slow as the network gets less reliable.

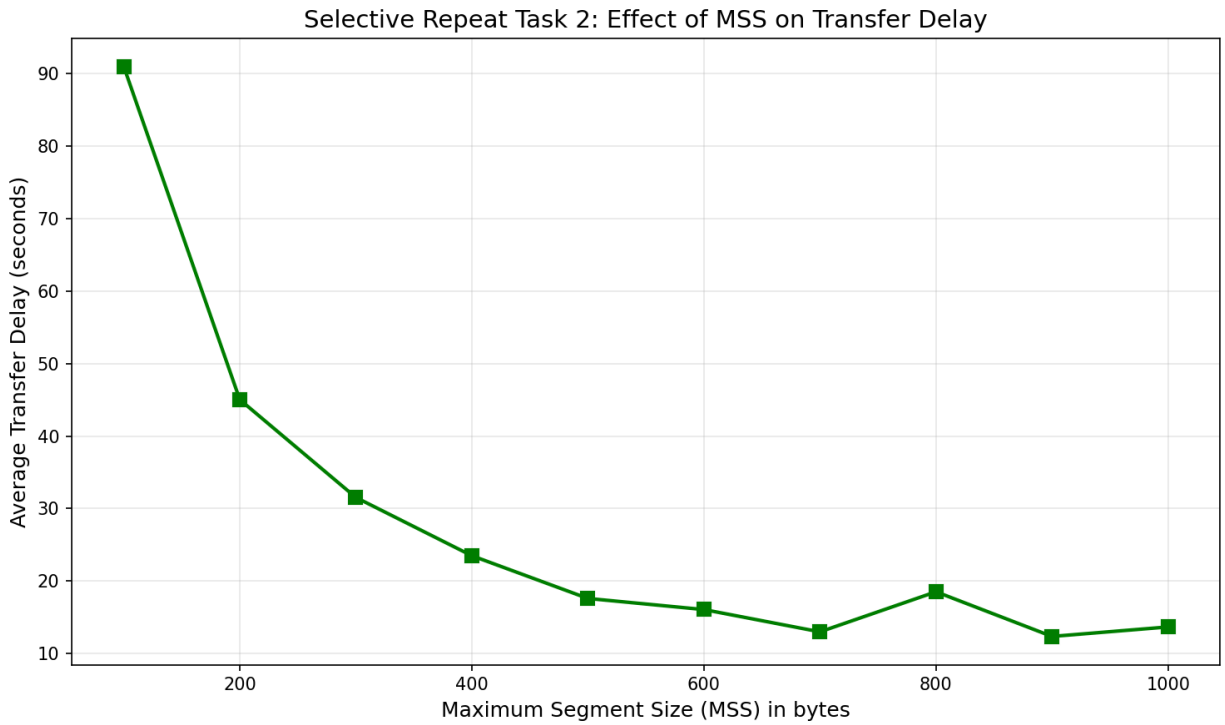
B. Selective Repeat ARQ Protocol

Task 1:



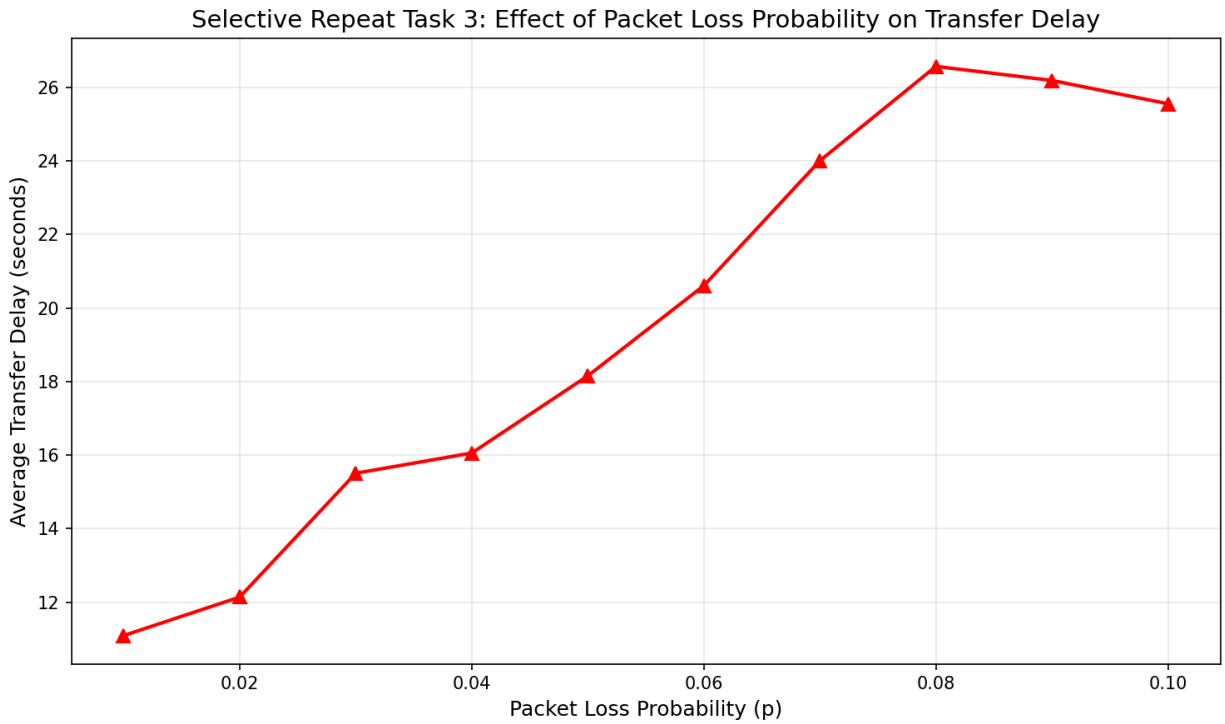
Interpretation: The plot shows shows delay falling hard as the window grows, from about 121.85 s at N=1 down to about 18.11 s at N=64. The expected pattern is: as the sender window grows, delay should drop sharply at first because SR can keep more packets outstanding and does not have to wait for each ACK before sending the next packet. That is the pipelining benefit of the protocol. Once the window is large enough to keep the path busy, the curve should flatten, because the bottleneck is no longer “not enough packets in flight” but the actual network/receiver/ACK turnaround. So the strong drop at small N and the leveling off later are exactly what SR should do. The small bumps at larger N are normal measurement variation, not a protocol contradiction.

Task 2:



Interpretation: In this plot larger MSS means fewer packets for the same file, so fewer data headers, fewer ACKs, fewer opportunities for loss, and fewer timeout events. In SR, that usually lowers delay because the transfer is driven by packet count as much as by bytes. The curve should therefore decrease as MSS increases, with diminishing returns once packet overhead becomes a smaller fraction of total cost. That is what the plot shows. Small non-monotonic points are plausible because the total time is still affected by random losses, timing jitter, and how many packets happen to be retransmitted in a given trial.

Task 3:



Interpretation: SR should worsen as loss increases, but not as badly as Go-Back-N. That is because Selective Repeat retransmits only the missing packets, not the whole window after a loss. So the expected shape is an upward trend with loss, but a gentler one than GBN would show. The plot matches that: delay rises as loss probability goes up, because more packets are missing and more individual timeouts fire. The slight flattening at the highest loss values is normal when you only average five trials; the underlying relationship is still “more loss, more retransmission, more delay.”